PermWorld: exact tasks, explicit relationships, auditable evidence

a A mathematical task resource, extended

Endpoint names, templates and instances are different counts

Original PermWorld

V2: 20 endpoints | V3: 20 | Property32: 32

Union: 48 endpoint names (overlapping catalogs)

4 upstream encodings; exact permutation tasks

Reusable platform v0.1

48 retained + 6 added = 54 endpoints

48 canonical templates; 62 declared instances

56 materialized unary instances; one-line input

Focus families	Templates / instances
Cycles	6 / 8
Subsequences	5 / 9
Blocks	5 / 7

Focus: 16 templates / 24 instances / 22 nonconstant.
Counts do not establish mathematical independence.

b Relationships are explicit design objects

74 relation records; typed evidence and query conditions

Transformed-input relation

$\mathrm{LIS}(p) = \mathrm{LDS}(C(p))$

The source answer uses a complemented input.

Joint answers on the same input

$\mathrm{FP}(p) = n - \mathrm{exceedances}(p) - \mathrm{deficiencies}(p)$

Formula availability does not establish model use.

Exact generators → full-orbit isolated data

Paired training → adaptation → audited archive

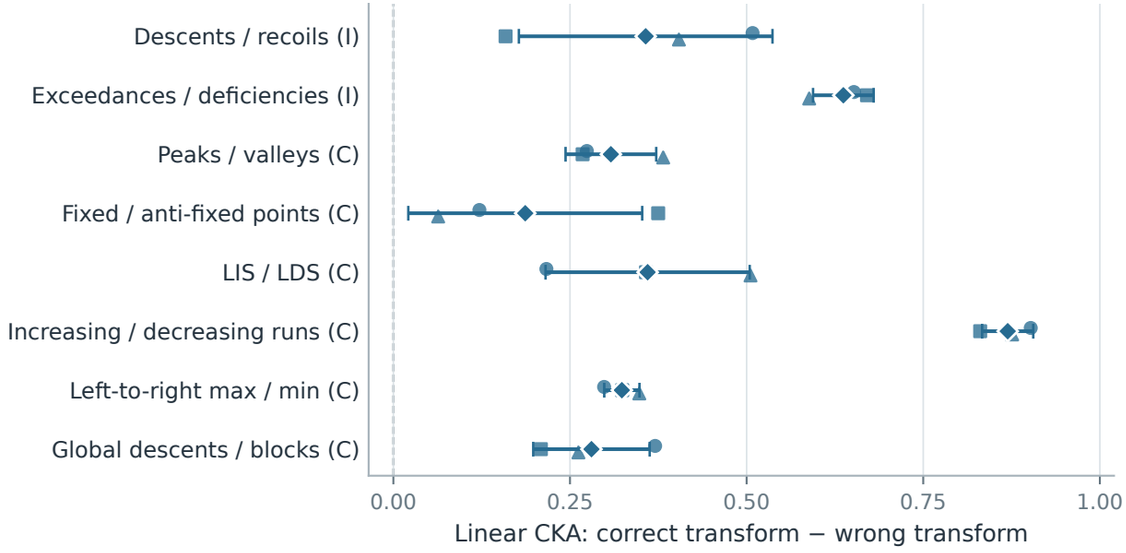
Versioned complete-answer + EOS evaluation

Input isolation and legal task relations are separate checks.

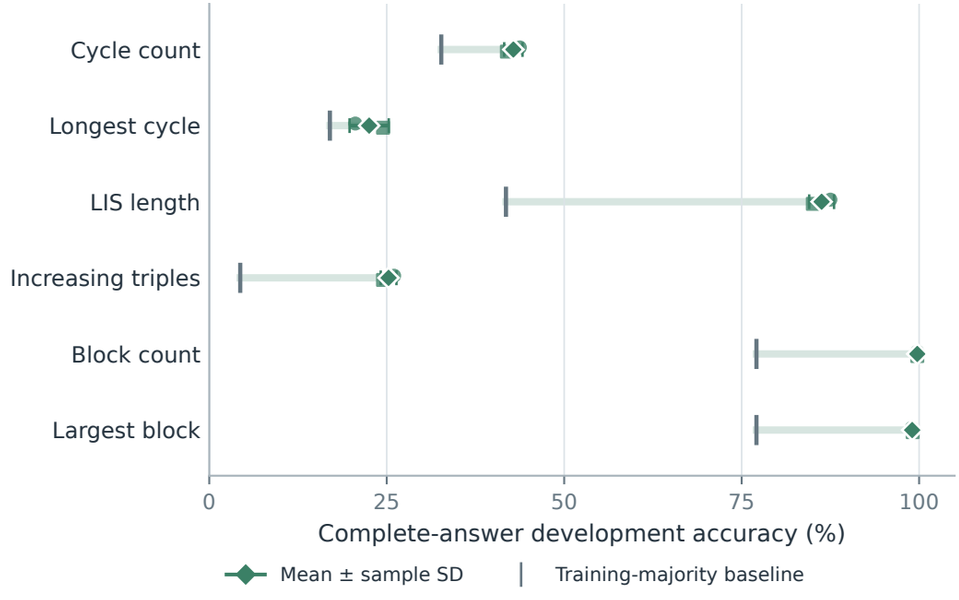
Aliases and constants remain visible.
Missing relations mean unclassified, not unrelated.

c Original relations leave a geometric signature

One-line · task-free final_norm · 3 seeds per relation
4096 shared validation inputs, pooled over n=2–30



I = inverse; C = complement. Seed variation is descriptive;
8 relation-level contrasts, not 24 independent relations. Geometry ≠ transfer.

d Learnability is measured, not assumed

Platform calibration · dev endpoint · 2 seeds · n=8,10,12